

Exponents, Radicals & Logarithms

Diagnostic and benchmark teacher companion

DOMAIN → NORMALIZE → REPRESENT → TRANSFORM → CHECK

Teacher-only companion · independent answer and misconception diagnostics

Teacher Diagnostic Key — Exponents, Radicals and Logarithms

Recognition and First-Line Lab

1. B. 8 and 4 are powers of 2, so common-base normalization is cheaper than logarithms.
2. C. Principal square root gives $\sqrt{x^2} = |x|$.
3. C. $\sqrt{U}$ requires $U \geq 0$; to make squaring $\sqrt{U} = V$ reversible, also require $V \geq 0$.
4. B. The conjugate converts the denominator to $7 - 5$.
5. B. The inner radical remains part of the outer radicand.
6. B. Without the sign condition, squaring may create candidates; check the original equation.
7. B. Base $x - 2 > 0$, base $x - 2 \neq 1$, argument $x + 1 > 0$.
8. B. Set $t = \log_a b$; then $\log_b a = 1/t$.
9. $3^{3x} = 3^{2x+2}$.
10. $5 - 2x \geq 0$, so $x \leq 5/2$.
11. $2x + 3 \geq 0$ and $x + 1 \geq 0$; together the reversible-square domain is $x \geq -1$.
12. Multiply by $(\sqrt{11} - \sqrt{3})/(\sqrt{11} - \sqrt{3})$.
13. $m + n = 13$ and $mn = 40$.
14. $x + 6 \geq 0$ and $x - \sqrt{x + 6} \geq 0$.
15. $x > 0$, $x \neq 1$, and exponent form $(x)^4 = 16$.
16. Set $t = \log_a b > 0$; then $t + 6/t = 5$.

Practice and Transfer Bank

1. $27^{2/3} = (3^3)^{2/3} = 3^2 = 9$.
2. $3^{2x} = 3^3$, so $x = 3/2$.
3. $\sqrt{200} = \sqrt{100 \cdot 2} = 10\sqrt{2}$.
4. $5 - 2x \geq 0$, so $x \leq 5/2$.
5. $(\sqrt{7} + \sqrt{5})/(7 - 5) = (\sqrt{7} + \sqrt{5})/2$.
6. Domain requires $x \geq 0$. Squaring gives $x + 6 = x^2$, hence $x = 3, -2$; only 3 is valid.
7. $|x - 4| = 3$, so $x = 1, 7$.
8. $(2 + \sqrt{3})^2 = 7 + 4\sqrt{3}$, so the expression is $2 + \sqrt{3}$.
9. Domain gives $x + 1 \geq 0$. Squaring gives $2x + 3 = (x + 1)^2$, hence $x^2 = 2$. Only $x = \sqrt{2}$ satisfies $x + 1 \geq 0$.
10. Let $A = \sqrt{x + 1}$, $B = \sqrt{x - 3}$. Then $A - B = 1$ and $(A - B)(A + B) = 4$, so $A + B = 4$. Hence $A = 5/2$, $B = 3/2$, giving $x = 21/4$.
11. $13 - 4\sqrt{10} = (\sqrt{8} - \sqrt{5})^2$; since $\sqrt{8} > \sqrt{5}$, the value is $\sqrt{8} - \sqrt{5} = 2\sqrt{2} - \sqrt{5}$.
12. Squaring once gives $\sqrt{x + 6} = x - 4$, so $x \geq 4$. Squaring again gives $x^2 - 9x + 10 = 0$; only $x = (9 + \sqrt{41})/2$ satisfies the sign condition.
13. $x - 1 = 8$, so $x = 9$.
14. Set $t = \log_a b > 0$. Then $t + 1/t = 5/2$, so $t = 2$ or $1/2$. Thus $b = a^2$ or $a = b^2$. Squares within 20 come from bases 2, 3, 4, giving 3 pairs each direction, total 6.
15. $x = \log_2 7$.
16. Base conditions: $x > 0$, $x \neq 1$. Exponent form gives $x^4 = 16$; the legal real base is $x = 2$.
17. 54. Set $t = \log_a b$; then $t + 6/t = 5$, so $t = 2, 3$. Count $b = a^2$ for $a = 2..44$ (43 pairs) and $b = a^3$ for $a = 2..12$ (11 pairs).
18. 91. Domain/sign analysis forces $y = 0$; then $\sqrt{x + a} = x - a$. Set $t = x - a \geq 0$; this gives $a = t(t - 1)/2$. The largest admissible nonsquare below 100 is $t = 14 \rightarrow a = 91$.
19. Squaring gives $x = x^2 - 4x + 4$, so candidates are 1, 4. The original equation requires $x - 2 \geq 0$; therefore only $x = 4$ is valid.
20. $\log_2 x$ must be defined and non-negative. Squaring gives $\log_2 x = 4$, hence $x = 16$, which satisfies all domain conditions.

Mixed Mastery Test

- $2^{4x-4} = 2^{6x+3}$, so $x = -7/2$.
- Domain requires $x \geq 0$. Squaring gives $x^2 - 3x + 2 = 0$, so $x = 1, 2$; both satisfy the original equation.
- Since $2 - \sqrt{3} > 0$, $\sqrt{(2 - \sqrt{3})^2} = 2 - \sqrt{3}$.
- $3(\sqrt{5} - 1)/(5 - 1) = 3(\sqrt{5} - 1)/4$.
- Let $A = \sqrt{x+5}$, $B = \sqrt{x+1}$. Then $A - B = 1$, $A + B = 4$; hence $A = 5/2$, giving $x = 5/4$.
- $(2 + \sqrt{5})^2 = 9 + 4\sqrt{5}$, so the value is $2 + \sqrt{5}$.
- First square: $\sqrt{x+2} = x - 1$, so $x \geq 1$. Second square gives $x^2 - 3x - 1 = 0$; valid solution $x = (3 + \sqrt{13})/2$.
- Candidates after squaring are 1, 4; original sign condition $x - 2 \geq 0$ leaves 4.
- $x - 1 = 9$, so $x = 10$.
- Base $x > 0$, $x \neq 1$; exponent form $x^3 = 27$ gives $x = 3$.
- Set $t = \log_a b > 0$. Then $t + 2/t = 3$, so $t = 1$ or 2. $t = 1$ gives $a = b$: 99 pairs. $t = 2$ gives $b = a^2$ for $a = 2..10$: 9 more. Total 108.
- Squaring is reversible because both sides are non-negative. The same reduction gives $a = t(t - 1)/2$. Under $a < 50$, $t = 10$ gives 45 and $t = 11$ gives 55; 45 is nonsquare. Answer 45.

Historical anchor audit

IOQM-2023-Q02 = 54. - logarithm bases/arguments are legal because $a, b \geq 2$; - $t = \log_a b > 0$ and reciprocal log is $1/t$; - quadratic roots are 2, 3; - integer bounds give $43 + 11 = 54$ ordered pairs.

IOQM-2025-Q28 = 91. - exact controlled source contains the nested radical $\sqrt{x - \sqrt{x + a}}$; - principal-root signs are recorded before squaring; - irrationality/integer structure forces $y = 0$; - all squaring steps used in the final route are reversible under recorded signs; - $a = t(t - 1)/2$, and $t = 14$ gives the maximal admissible nonsquare 91.

Diagnostic map

- $\sqrt{x^2}$ replaced by $x \rightarrow$ restore principal-root rule $|x|$;
- radical squared with no sign record $\rightarrow$ classify the move as candidate-generating and check the original;
- denominator rationalized mechanically $\rightarrow$ ask what difference-of-squares structure the conjugate exposes;
- nested radical flattened $\rightarrow$ preserve inner radical as one object;
- logarithm manipulated before base/argument domain $\rightarrow$ return to domain conditions;
- logarithm introduced despite an obvious common base $\rightarrow$ normalize first;
- quadratic roots all accepted after squaring $\rightarrow$ test sign conditions and original equation;
- integer search started before algebraic structure $\rightarrow$ solve the real/domain problem first, then filter integers.

Teacher Key — ALG-06 Benchmark Assimilation Lab

This key supports 07_Benchmark_Assimilation_Lab.md. It is separate from the main Teacher Diagnostic Key so the benchmark-specific learner diagnostic can be audited independently.

A. RECONNECT

1. $\sqrt{x^2} = |x|$ over the reals.
2. $2x - 1 \geq 0$ for the radicand and $x - 2 \geq 0$ because the left side is non-negative. Together $x \geq 2$; on this restricted domain squaring is reversible.
3. Common base. Write $8 = 2^3$, $4 = 2^2$ before considering logarithms.
4. Conjugate $\sqrt{7} + \sqrt{3}$; the product is $7 - 3 = 4$, a rational denominator.
5. For $\log_{x-1}(x+2)$: $x - 1 > 0$, $x - 1 \neq 1$, $x + 2 > 0$. Thus $x > 1$ and $x \neq 2$; the argument condition is then automatic.
6. Square root is not linear over subtraction. The entire expression $x - \sqrt{x+a}$ is the radicand of the outer root and must remain one object.
7. $\log_b a = 1/t$, provided both logarithms are defined and $t \neq 0$.
8. Check every candidate against the original domain/sign restrictions and the original unsquared equation.

B. Error laboratory

Error 1

Correct identity: $\sqrt{(x-5)^2} = |x-5|$. It equals $x-5$ only when $x \geq 5$.

Error 2

Before squaring $\sqrt{x+3} = 1-x$, require $x+3 \geq 0$ and $1-x \geq 0$. Thus $-3 \leq x \leq 1$. Squaring is reversible only on that restricted domain; any algebraic candidate outside it is invalid.

Error 3

$\sqrt{x - \sqrt{x+a}}$ cannot be distributed over subtraction. Preserve the outer radicand; first require $x+a \geq 0$ and $x - \sqrt{x+a} \geq 0$.

Error 4

For $\log_{x-2}(x+1)$, first require $x-2 > 0$, $x-2 \neq 1$, $x+1 > 0$. Only then convert to exponent form.

Error 5

$9^x = 27$ is already $3^{2x} = 3^3$, so compare exponents: $2x = 3$. Logs are valid but inefficient.

Error 6

First solve/reduce the radical/logarithmic structure under its real-domain conditions. Apply integrality only after the admissible algebraic family is known.

C. ADOPT

1. Write $3x + 4 \geq 0$ and, because $\sqrt{3x+4} = x$, also $x \geq 0$; then square on $x \geq 0$.
2. Multiply numerator and denominator by $\sqrt{13} + \sqrt{5}$.
3. Test $\sqrt{m} + \sqrt{n}$: $m + n = 17$, $mn = 60$ (hence $\{m, n\} = \{12, 5\}$).
4. Domain: $x - 1 > 0$, $x - 1 \neq 1$; then $(x - 1)^3 = 27$.
5. Set $t = \log_a b > 0$, so $\log_b a = 1/t$; obtain $t + 2/t = 3$.
6. Rewrite in base 2: $2^{4x-4} = 2^{3x+6}$.
7. Record $x + 6 \geq 0$ and $x - \sqrt{x+6} \geq 0$; after the first justified square, write $x - \sqrt{x+6} = 4$.
8. Require $x - 4 \geq 0$ before the second square; this is the missing sign condition that makes the next square reversible.

D. TRANSFER

1. Yes. Compare $11 - 6\sqrt{2}$ with $(\sqrt{m} - \sqrt{n})^2 = m + n - 2\sqrt{mn}$. Need $m + n = 11$, $mn = 18$; $\{m, n\} = \{9, 2\}$, so $u = 3 - \sqrt{2}$ because $u \geq 0$.
2. Set $t = \log_a b > 0$; then $t + 1/t = 5/2$, so $2t^2 - 5t + 2 = 0$, giving $t = 2$ or $1/2$. Thus $b = a^2$ or $a = b^2$; the bounds reduce the problem to counting integer square pairs.
3. $\sqrt{P(x)}$ is non-negative, so any solution also needs $Q(x) \geq 0$. Squaring may solve $P(x) = Q(x)^2$, but candidates with $Q(x) < 0$ or invalid radicand must be rejected in the original equation.
4. Separate rational and irrational parts. Since $\sqrt{a}$ is irrational for nonsquare a , an equality between an integer/rational quantity and $k + m\sqrt{a}$ forces the irrational coefficient to satisfy the necessary cancellation condition (often $m = 0$ or another exact coefficient equation).
5. Normalize both sides to powers of 6 and compare exponents. Logs would preserve correctness but introduce an unnecessary operation and extra notation without reducing the structure further.
6. Example: $\sqrt{x+4} = x+1$ together with the recorded condition $x \geq -1$ and $x+4 \geq 0$. On that domain both sides are non-negative, so squaring is an equivalence.

E. Six-question assimilation test — diagnostic rubric

A strong response should contain all six elements:

1. a specific visible clue, not a chapter name;
2. the invariant/domain reason the proposed method is valid;
3. a reusable recognition cue;
4. a genuine near-neighbour requiring a different route;
5. two mathematically useful opening lines;
6. a changed-surface example preserving the invariant, with a plausible route.

Diagnostic tags:

- recognition: clue not identified;
- representation: wrong route chosen despite a cheaper one;
- domain_condition: sign/base/argument condition omitted;
- equivalence: implication-only transformation treated as reversible;
- execution: correct route but algebra error;
- checking: candidates not returned to original equation;
- transfer: disguised version changes the mathematics rather than only the surface.

Independent QA disposition

All deterministic answers and route claims above were recomputed after the learner lab was authored. No answer is inherited from the learner-facing document.

BENCHMARK_LAB_KEY_STATIC_CHECK = PASS.